

Unit vector flows

OPG-136	L1 - Tractable	Ouvert	Graph Theory
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Énoncé original (anglais, reproduction mot pour mot)

Conjecture For every graph G without a bridge, there is a flow $\phi: E(G) \rightarrow S^2 = \{x \in \mathbb{R}^3: |x| = 1\}$.

Conjecture There exists a map $q: S^2 \rightarrow \{-4, -3, -2, -1, 1, 2, 3, 4\}$ so that antipodal points of S^2 receive opposite values, and so that any three points which are equidistant on a great circle have values which sum to zero.

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